

DISCRETE MATHEMATICS

FINAL TEST

Date: _____

Name: _____

Three exercises, **5 points** each (total: **15 points**). Justify all answers. Time: **90 minutes**. _____

Exercise 1

(5 points) Generating Functions

[2 pts] Use generating functions to find a closed-form expression for a_n , where

$$a_n = 4a_{n-1} - 3a_{n-2}, \quad n \geq 2, \quad a_0 = 1, \quad a_1 = 5.$$

Write the generating function $F(x) = \sum_{n=0}^{\infty} a_n x^n$, solve for $F(x)$, and find a_n .

[3 pts] Using generating functions, find the number of non-negative integer solutions to

$$e_1 + e_2 + e_3 = 12$$

subject to: e_1 is odd, $0 \leq e_2 \leq 4$, $e_3 \geq 0$.

Write the product of generating functions corresponding to each variable, simplify, and extract the coefficient of x^{12} .

Exercise 2

(5 points) Coding Theory

All codes are binary (over $\mathbb{F}_2$). Consider the linear code $\mathcal{C}$ with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

[1 pt] State n and k . Write the parity-check matrix H using the relation $G = [I_k \mid P]$.

[1 pt] List all 2^k codewords of $\mathcal{C}$.

[1 pt] Determine $d_{\min}(\mathcal{C})$. How many errors does $\mathcal{C}$ detect? How many does it correct? Is $\mathcal{C}$ a perfect code? (Recall: $\mathcal{C}$ is perfect if $2^k \cdot V(n, t) = 2^n$, where $V(n, t) = \sum_{i=0}^t \binom{n}{i}$.)

[2 pts] The word $\mathbf{r} = (1, 1, 0, 0, 0, 1, 0)$ was received. Assuming at most one error:

- (i) Compute the syndrome $\mathbf{s} = H\mathbf{r}^\top$.
- (ii) Identify the error position by comparing $\mathbf{s}$ with the columns of H .
- (iii) Write the corrected codeword and recover the original message $\mathbf{u} \in \mathbb{F}_2^3$.

Exercise 3

(5 points) Graph Theory

Definitions. An *Eulerian circuit* in a connected graph G is a closed walk that traverses every edge of G exactly once. An *Eulerian path* is a walk (not necessarily closed) that traverses every edge exactly once.

[3 pts] Use the Erdős–Gallai theorem to determine which of the following sequences is the degree sequence of a simple graph. For any sequence that is graphical, construct a realizing graph. For any that is not, state which condition fails.

(i) $d_1 = (5, 4, 3, 3, 3, 2)$

(ii) $d_2 = (5, 4, 4, 3, 1, 1)$

[2 pts] Let G be any simple graph whose degree sequence is d_1 . Using only the degree sequence (without constructing G explicitly), determine whether G has an Eulerian circuit, an Eulerian path (but not a circuit), or neither. Prove your answer.

DISCRETE MATHEMATICS

FINAL TEST — SOLUTIONS

Exercise 1

(5 points) Generating Functions

[2 pts] Recurrence $a_n = 4a_{n-1} - 3a_{n-2}$, $a_0 = 1$, $a_1 = 5$

Let $F(x) = \sum_{n=0}^{\infty} a_n x^n$. Multiplying the recurrence by x^n and summing from $n = 2$ to ∞ :

$$(F - a_0 - a_1 x) = 4x(F - a_0) - 3x^2 F.$$

Substituting $a_0 = 1$, $a_1 = 5$ and collecting terms in F :

$$F(1 - 4x + 3x^2) = 1 + 5x - 4x = 1 + x.$$

The denominator factors as $(1 - x)(1 - 3x)$, so

$$F(x) = \frac{1 + x}{(1 - x)(1 - 3x)}.$$

Partial fractions: write $\frac{1 + x}{(1 - x)(1 - 3x)} = \frac{A}{1 - x} + \frac{B}{1 - 3x}$. Then $1 + x = A(1 - 3x) + B(1 - x)$. Setting $x = 1$: $A = -1$; setting $x = \frac{1}{3}$: $B = 2$. Hence

$$F(x) = \frac{-1}{1 - x} + \frac{2}{1 - 3x},$$

and extracting coefficients:

$$\boxed{a_n = 2 \cdot 3^n - 1.}$$

Check. $a_0 = 1✓$, $a_1 = 5✓$, $a_2 = 2 \cdot 9 - 1 = 17 = 4 \cdot 5 - 3 \cdot 1✓$.

[3 pts] Counting solutions to $e_1 + e_2 + e_3 = 12$

The constraints give the following generating functions for each variable:

$$e_1 \text{ odd : } \frac{x}{1 - x^2}, \quad 0 \leq e_2 \leq 4 : \frac{1 - x^5}{1 - x}, \quad e_3 \geq 0 : \frac{1}{1 - x}.$$

The product is

$$F(x) = \frac{x}{1 - x^2} \cdot \frac{1 - x^5}{1 - x} \cdot \frac{1}{1 - x} = \frac{x(1 - x^5)}{(1 - x)^2(1 - x^2)} = \frac{x(1 - x^5)}{(1 - x)^3(1 + x)}.$$

The required count is $[x^{12}]F(x) = [x^{11}]\frac{1 - x^5}{(1 - x)^3(1 + x)}$, which splits as

$$[x^{11}]\frac{1}{(1 - x)^3(1 + x)} - [x^6]\frac{1}{(1 - x)^3(1 + x)}.$$

To find $[x^n] \frac{1}{(1-x)^3(1+x)}$, decompose into partial fractions:

$$\frac{1}{(1-x)^3(1+x)} = \frac{1/2}{(1-x)^3} + \frac{1/4}{(1-x)^2} + \frac{1/8}{1-x} + \frac{1/8}{1+x}.$$

Extracting coefficients and simplifying:

$$[x^n] \frac{1}{(1-x)^3(1+x)} = \frac{1}{2} \binom{n+2}{2} + \frac{n+1}{4} + \frac{1}{8} + \frac{(-1)^n}{8}.$$

This gives, for odd n : $\frac{(n+1)(n+3)}{4}$; for even n : $\frac{(n+2)^2}{4}$.

Applying to our case:

$$[x^{11}] \frac{1}{(1-x)^3(1+x)} = \frac{12 \cdot 14}{4} = 42, \quad [x^6] \frac{1}{(1-x)^3(1+x)} = \frac{8^2}{4} = 16.$$

Number of solutions = $42 - 16 = 26$.
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Exercise 2

(5 points) Coding Theory

[1 pt] Parameters and parity-check matrix

$G = [I_3 \mid P]$ with $n = 7$, $k = 3$, $n - k = 4$, and $P = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$. The parity-check matrix is $H = [P^\top \mid I_4]$:

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

[1 pt] Codewords

$$\begin{aligned} (0, 0, 0) &\rightarrow (0, 0, 0, 0, 0, 0, 0) \\ (1, 0, 0) &\rightarrow (1, 0, 0, 1, 0, 1, 1) \\ (0, 1, 0) &\rightarrow (0, 1, 0, 1, 1, 0, 1) \\ (0, 0, 1) &\rightarrow (0, 0, 1, 0, 1, 1, 1) \\ (1, 1, 0) &\rightarrow (1, 1, 0, 0, 1, 1, 0) \\ (1, 0, 1) &\rightarrow (1, 0, 1, 1, 1, 0, 0) \\ (0, 1, 1) &\rightarrow (0, 1, 1, 1, 0, 1, 0) \\ (1, 1, 1) &\rightarrow (1, 1, 1, 0, 0, 0, 1) \end{aligned}$$

[1 pt] $d_{\min}$, detection, correction, perfectness

Every nonzero codeword has weight 4, so $d_{\min} = 4$. The code detects up to $d_{\min} - 1 = 3$ errors and corrects up to $t = \lfloor 3/2 \rfloor = 1$ error.

Perfect code check. $V(7, 1) = \binom{7}{0} + \binom{7}{1} = 1 + 7 = 8$. $2^k \cdot V(7, 1) = 8 \cdot 8 = 64 \neq 128 = 2^7$. The code is **not perfect**.

[2 pts] Syndrome decoding of $\mathbf{r} = (1, 1, 0, 0, 0, 1, 0)$

(i) Syndrome.

$$\mathbf{s} = H\mathbf{r}^\top = \begin{pmatrix} 1+1 \\ 1 \\ 1+1 \\ 1+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

(ii) **Error position.** The columns of H are $\mathbf{h}_1 = (1, 0, 1, 1)^\top$, $\mathbf{h}_2 = (1, 1, 0, 1)^\top$, $\mathbf{h}_3 = (0, 1, 1, 1)^\top$, $\mathbf{h}_4 = (1, 0, 0, 0)^\top$, $\mathbf{h}_5 = (0, 1, 0, 0)^\top$, $\mathbf{h}_6 = (0, 0, 1, 0)^\top$, $\mathbf{h}_7 = (0, 0, 0, 1)^\top$. Since $\mathbf{s} = (0, 1, 0, 0)^\top = \mathbf{h}_5$, the error is at **position 5**.

(iii) **Corrected codeword and message.** Flip bit 5:

$$\hat{\mathbf{c}} = (1, 1, 0, 0, \mathbf{1}, 1, 0).$$

Since $G = [I_3 \mid P]$, the message is the first $k = 3$ bits:

$$\boxed{\mathbf{u} = (1, 1, 0)}.$$

Verification. $(1, 1, 0) G = \mathbf{g}_1 + \mathbf{g}_2 = (1, 1, 0, 0, 1, 1, 0) = \hat{\mathbf{c}} \checkmark$.

Exercise 3

(5 points) Graph Theory

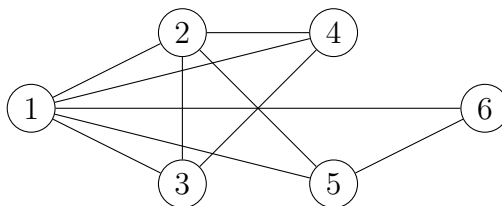
[3 pts] Erdős–Gallai theorem

Sequence $d_1 = (5, 4, 3, 3, 3, 2)$. The sum is $5 + 4 + 3 + 3 + 3 + 2 = 20$ (even✓). The

Erdős–Gallai condition $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$ is verified for each k :

k	$\sum_{i=1}^k d_i$	$k(k-1) + \sum_{i>k} \min(d_i, k)$	satisfied?
1	5	$0 + 1 + 1 + 1 + 1 + 1 = 5$	✓
2	9	$2 + 2 + 2 + 2 + 2 = 10$	✓
3	12	$6 + 3 + 3 + 2 = 14$	✓
4	15	$12 + 3 + 2 = 17$	✓
5	18	$20 + 2 = 22$	✓
6	20	30	✓

d_1 is **graphical**. A realizing graph: vertex 1 has degree 5 (connected to all others); vertex 2 additionally connects to 3, 4, 5; vertex 3 connects to 4; vertex 5 connects to 6.



Degrees: $\deg(1) = 5$, $\deg(2) = 4$, $\deg(3) = 3$, $\deg(4) = 3$, $\deg(5) = 3$, $\deg(6) = 2$ ✓.

Sequence $d_2 = (5, 4, 4, 3, 1, 1)$. The sum is 18 (even✓). Check $k = 2$: $\sum_{i=1}^2 d_i = 9$, but $k(k-1) + \sum_{i>2} \min(d_i, k) = 2 + 2 + 2 + 1 + 1 = 8$. Since $9 > 8$, the condition **fails at** $k = 2$. The sequence d_2 is **not graphical**.

[2 pts] Eulerian properties of any realization of d_1

Claim. No graph realizing d_1 has an Eulerian circuit or an Eulerian path.

Proof — no Eulerian circuit. Suppose G has an Eulerian circuit C . Consider any vertex v . Each time C passes through v , it arrives along one edge and departs along another, consuming a pair of edges incident to v . Since C is a closed walk that uses every edge exactly once, the edges at every vertex are entirely accounted for by such pairs. Therefore every vertex must have *even* degree. However, d_1 contains the degrees 5, 3, 3, 3, which are odd, so no such circuit exists.

Proof — no Eulerian path. Suppose G has an Eulerian path from u to v (with $u \neq v$). By the same pairing argument, every vertex *except* u and v has its incident edges

consumed in pairs, hence has even degree. At u , the path departs without a corresponding arrival (the first step), so exactly one edge at u is unmatched, forcing $\deg(u)$ to be odd. By the same reasoning $\deg(v)$ is odd. Thus an Eulerian path forces *exactly two* vertices to have odd degree.

In any realization of $d_1 = (5, 4, 3, 3, 3, 2)$, exactly the vertices of degrees 5, 3, 3, 3 have odd degree, giving **four** vertices of odd degree. Since $4 \neq 2$ and $4 \neq 0$, neither an Eulerian circuit nor an Eulerian path exists. $\square$